

DATA SCIENCE & INDUSTRIAL ANALYTICS

AI4I 2020 Predictive Maintenance

Exploratory Data Analysis, Quality Assessment & Predictive Modeling Insights

📊 10,000 Machine Observations

📋 14 Industrial Features

🛡️ Data Quality Assessment

Executive Summary & Key Metrics

Dataset Snapshot

The AI4I 2020 dataset simulates real-world mill machinery operation parameters to enable machine learning for predictive maintenance.

Our comprehensive audit identified key failure distributions, physical parameter relationships, and specific target label inconsistencies that require handling prior to modeling.

10,000

Total Instances

Complete records with zero missing values

3.39%

Overall Failure Rate

339 target machine failures recorded

5

Failure Subtypes

HDF, OSF, PWF, TWF, and RNF modes

9

Label Noise Cases

Sub-type failures marked target = 0

Dataset Overview & Schema Structure



Machine Metadata

Includes unique record IDs (UID 1–10,000) and Product ID categorical codes categorizing variants into Low, Medium, and High quality classes.



Process Parameters

Real-time physical sensors measuring Air Temp (K), Process Temp (K), Rotational Speed (RPM), Torque (Nm), and Tool Wear duration (minutes).



Target & Modes

Binary Machine Failure flag paired with 5 specific subtype indicators reflecting physical constraint violations and random breakdowns.

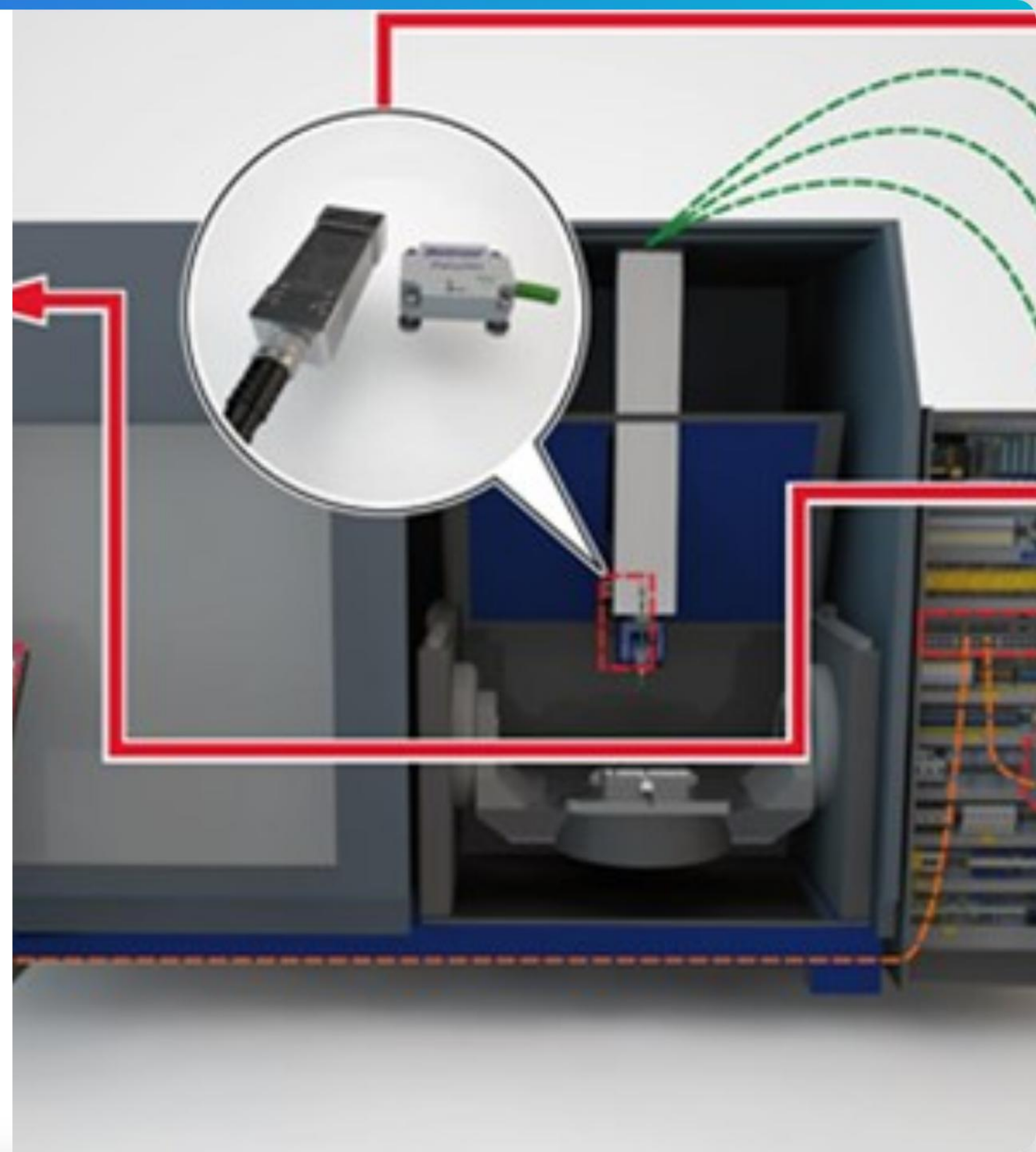
Operational Parameters Analysis

Physical Operating Envelopes

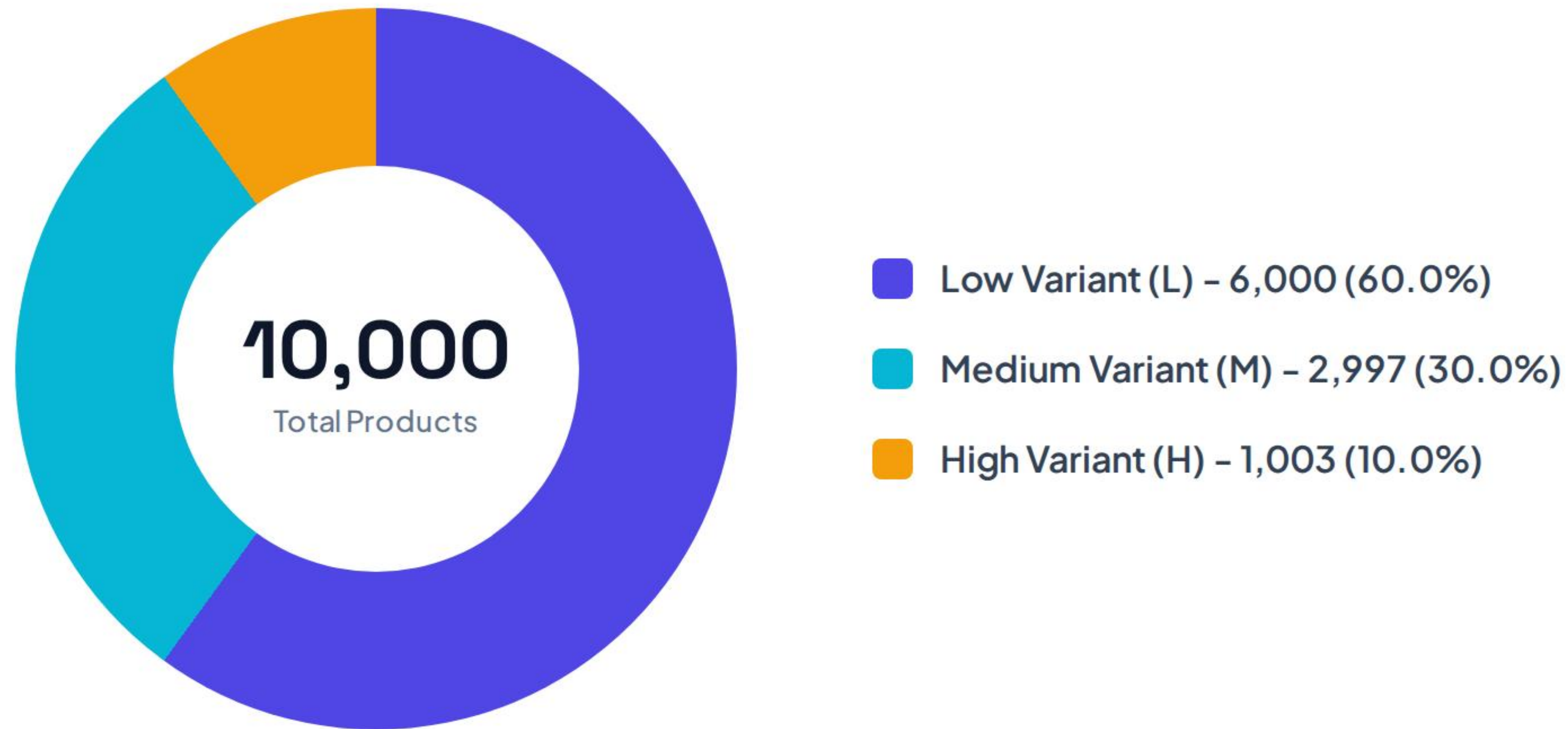
Industrial sensors capture vital continuous operational metrics. Operational parameters adhere strictly to known physical relationships.

Thermal: Air temperature averages ~300 K (sd=2K) while Process temperature averages ~310 K (sd=1K).

Mechanical: Rotational Speed (1538 RPM avg) and Torque (40 Nm avg) exhibit strong inverse physical power correlation ($\text{Power} = \text{Torque} \times \omega$).



Product Quality Variant Distribution



Product quality variants affect tool wear accumulation rates (+2 min for L, +3 min for M, +5 min for H) and overstrain failure limits.

Failure Mode Frequency Breakdown



Heat Dissipation and Overstrain account for over 57% of specific failure causes. Subtype total count (372) exceeds unique target failures (339) due to multi-failure co-occurrences.

Data Quality & Target Inconsistencies

✖ Target Label Discrepancies

9 Records: Subtype failure flags are set to 1 while Machine Failure target is 0 (e.g., 5 Random, 2 Tool Wear, 1 Power, 1 Heat Dissipation).

24 Records: Exhibit multiple simultaneous failure mode flags, demonstrating compound mechanical stress during operation.

📈 Skewness & Outliers

Rotational Speed: Right-skewed (skew = 1.99) with extreme values up to 2,886 RPM creating long tails.

Torque: Normally distributed around 40 Nm with symmetric outliers at both low (<16.7 Nm) and high (>63.3 Nm) extremes.

Physical Sensor Correlation Insights

Key Feature Interactions

Understanding sensor physics improves predictive model engineering:

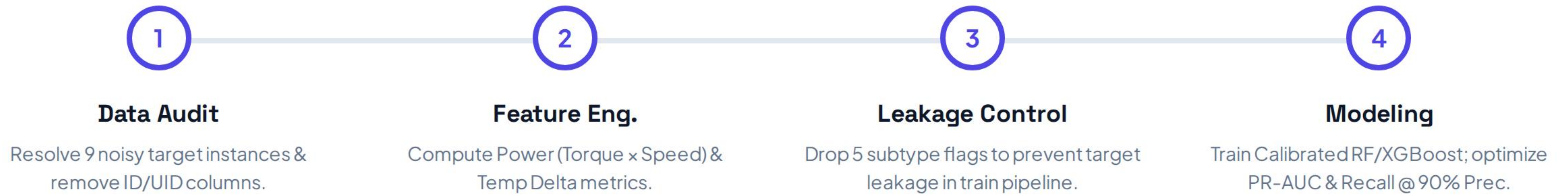
- **Thermal Delta:** High correlation ($r = 0.88$) between Air and Process Temperature; Delta < 8.6 K triggers HDF.
- **Power Thresholds:** Mechanical power output below 3,500 W or above 9,000 W causes Power Failure (PWF).
- **Stress Product:** Tool wear $\times$ Torque interaction triggers Overstrain Failure (OSF) when exceeding variant thresholds.



Statistical Feature Summary

Feature Name	Mean	Std Dev	Min	Median	Max	Skewness
Air Temperature [K]	300.00	2.00	295.30	300.10	304.50	-0.02
Process Temperature [K]	310.01	1.48	305.70	310.10	313.80	-0.01
Rotational Speed [rpm]	1538.78	179.28	1168.00	1503.00	2886.00	1.99
Torque [Nm]	39.99	9.97	3.80	40.10	76.60	-0.00
Tool Wear [min]	107.95	63.65	0.00	108.00	253.00	0.03

Machine Learning Implementation Roadmap



Strategic Engineering Recommendations



Target Leakage Protection: The 5 subtype failure columns (TWF, HDF, PWF, OSF, RNF) strictly define target criteria. They must be excluded prior to model training to avoid perfect leakage.



Domain Physics Feature Engineering: Synthesize explicit physical predictors—specifically Mechanical Power (W) and Thermal Gradient (K)—to help non-linear tree models isolate operating boundaries.



Imbalance Strategy: With a 3.39% failure rate, optimize models using Precision-Recall AUC (PR-AUC) and cost-weighted decision thresholds rather than raw accuracy.

Questions & Discussion

AI4I 2020 Predictive Maintenance Dataset Analysis Complete



Ready for Machine Learning Pipeline Integration

Image Sources



<https://static-01.extrica.com/articles/20788/20788-img2.jpg>

Source: www.extrica.com



<https://pytheasenergy.com/wp-content/uploads/2026/06/Real-Time-Well-Data-Analytics-1024x576.png>

Source: pytheasenergy.com